

Universidad Autónoma de Yucatán

Facultad de Matemáticas

Asignatura:

Programación Orientada a Objetos

Proyecto: UniRide

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1. Introduction & Objectives

1.1 Context and Problem Definition

Transportation is primarily one of the most frequent problems when commuting to the university. To address this, the government has implemented public transportation routes to mitigate this problem; this implies that students usually use this service, thereby reducing the burden to some extent.

However, the saturation of this medium typically increases during peak hours due to the high percentage of people who use it (not only students), which in turn causes overcrowding, resulting in certain routes operating at maximum passenger capacity. As a consequence, students face the possibility of being delayed. Another common problem related to this is that a large portion of students do not have a direct route or live in distant residential areas; for this reason, they have to take more than one route to reach their destination and, as a result, the student may potentially be delayed.

1.2 Project Justification

To solve these problems and make commuting and time management more efficient for certain students, a solution was developed consisting of a carpooling platform exclusive to the university community. Unlike a simple Facebook group or a message board, this functions with an automated system.

1.3 General Objective and Scope

Students (drivers and passengers) register their class schedules, the zone they leave from, which campus they are heading to, and how many minutes of wait-time tolerance they have. The system automatically cross-references all these variables and makes the perfect match, connecting those with compatible routes and schedules so they can share the ride and expenses safely.

2. System Specification

UniRide is a point-to-point shared transportation logistics platform designed exclusively for the university community. Its core function is not manual ride-hailing, but rather automated matchmaking.

The system takes data from two main actors (drivers and passengers) and evaluates a complex set of constraints (routes, class schedules, university, and acceptable wait times) to suggest the perfect connection, optimizing empty seats in vehicles and reducing students travel time.

2.1 Core Features:

2.1.1 Schedule Synchronization and Matching

The main differentiator is that this is not a simple, isolated trip; instead, students are provided with far greater flexibility. One of the core functionalities planned for implementation is the option to upload their academic course load, or to allow the user to configure it manually in case of errors or personal preference.

- **Driver:** Registers their home departure time and the end time of their last class.
- **Passenger:** Registers their arrival (entry) and departure (exit) times.
- **Logic Engine:** Compares both schedules. If the passenger leaves at 1:00 PM and the driver leaves at 1:15 PM, the system detects a viable overlap for the return trip.

2.1.2 Wait-Time Tolerance Filter

Since schedules do not always match exactly, users will be able to filter by the maximum amount of time they are willing to wait. This expands matching opportunities with other users.

- **Scenario A:** If a driver passes through a residential area at 6:30 AM, but the passenger prefers to leave at 6:45 AM and has configured a 20-minute tolerance, the system creates a Match (because 6:30 AM falls within the acceptable wait-time tolerance window).

- **Scenario B:** If the passenger is only willing to wait 5 minutes, the trip is either not scheduled or the system continues searching for matches that meet the given criteria.

2.1.3 Multi-University and Campus Filter

This filter allows the system to scale, ensuring it is not limited to a single institution or scope.

- A passenger from the Faculty of Mathematics can match with a driver from the Faculty of Engineering if both are located within the same campus or area (such as the Campus of Exact Sciences), optimizing the commute for students from neighboring faculties.

2.1.4 Dynamic Route Management

Both passengers and drivers establish route nodes. Potential integration of the Google Maps API to enhance the user experience and facilitate application usage.

2.1.5 Extra Features (Scalable):

- **Safe Ride Mode:** A boolean property or configuration setting that allows users to restrict their matches. For example, allowing female students to filter and ride exclusively with other registered women, or allowing a student to restrict their trips only to peers from the same faculty.
- **Reputation and Reliability System:** A behavior implemented via interfaces. If a driver cancels 10 minutes before the trip, or a passenger fails to show up at the meeting point, the system deducts "Reliability Points." If they fall below a certain threshold, the system triggers a user suspension.
- **Safe Spots:** Instead of locating the passenger's exact home address, the system defines a meeting point (well-known parks, libraries, convenience stores) so the driver does not have to deviate significantly from their main route.

Object-Oriented Architectural Design

UML Placeholder. (TO DO)

Class Dictionary (TO DO)

Modifier Acces	Element	Data Type/Return	Description

Estrategia de Persistencia y Errores

Description of what data will be read/written to the .txt file and the list of custom exceptions we will implement. (TO DO)

Justificación de Principios SOLID

Explain how the design respects extensibility (for example, how the Matchmaking interface complies with the Open/Closed principle). (TO DO)